

Power Query ETL Lab (Beginner) — Instructor Step-by-Step (Menu-Only)

Beginner workflow: no Power Query formulas / no M code. Every action uses buttons and menu options. Follow in order.

Files (keep in one folder): Raw_AP_Transactions.csv; Vendor_Alias_Map.csv; Vendor_Master.csv; Purchase_Orders.csv

Day 1 (50 minutes)	Day 2 (50 minutes)
Clean & standardize AP (types, vendor text, Amount, date, dept, PO, PO match status, duplicates, exceptions)	

Pre-class setup (2 minutes)

- 1 Unzip the 4 CSV files into a single folder (students must keep them together).
- 2 Open Excel → Blank workbook → Save as PQ_Beginner_Lab.xlsx.
- 3 Tell students: on first import always click Transform Data (not Load).

Day 1 — Clean the AP table

Goal: Create a clean, standardized AP query named AP_Raw (connection only).

1) Import Raw AP data

- 1 Excel ribbon → Data tab → Get Data (or From Text/CSV).
- 2 From File → From Text/CSV.
- 3 Select Raw_AP_Transactions.csv → Import.
- 4 In preview window click Transform Data (NOT Load).

2) Rename the query to AP_Raw

- 1 In Power Query Editor (left Queries pane), right-click the query name → Rename.
- 2 Type AP_Raw and press Enter.

3) Set key columns to Text first (prevents type errors)

- Click InvoiceDate column → Transform → Data Type → Text.
- Click Amount column → Transform → Data Type → Text.
- Click Vendor column → Transform → Data Type → Text.
- Recommended: also set Dept, PaymentTerms, PO_Number to Text.

4) Clean Vendor text (Trim + Clean)

- 1 Click Vendor column.
- 2 Transform → Format → Trim.
- 3 Transform → Format → Clean.

5) Amount cleanup + convert to number

Do the Replace steps ONLY if you see \$ signs or commas. If Amount already looks like plain numbers, skip to conversion.

- If needed: Amount → Transform → Replace Values → find \$ replace with blank → OK.
- If needed: Amount → Replace Values → find , replace with blank → OK.
- Convert: Amount → Transform → Data Type → Decimal Number (Replace current if prompted).

6) Convert InvoiceDate to Date

- InvoiceDate → Transform → Data Type → Date.
- If errors: Transform → Data Type → Using Locale... → Date; English (United States) → OK.

7) Standardize Dept (Replace Values)

- Dept column selected.
- Replace Values: Finance → FIN.
- Replace Values: Fin → FIN.
- Replace Values: Operations → OPS.
- Replace Values: Ops → OPS.
- Replace Values: Warehouse → WH.

8) Standardize PaymentTerms (Replace Values)

- PaymentTerms column selected.
- Replace Values: NET30 → Net 30.
- Replace Values: Net30 → Net 30.
- Replace Values: NET15 → Net 15.
- Replace Values: NET45 → Net 45.
- Replace Values: Net45 → Net 45.
- Replace Values: DOR → Due on receipt.

9) Day 1 checkpoint

- InvoiceDate shows real dates (not text).
- Amount is numeric (Decimal Number).
- Dept values are mostly FIN / OPS / IT / WH.
- PaymentTerms values are standardized (Net 30 / Net 15 / Net 45 / Due on receipt).

10) Save work without loading tables (connection only)

- 1 Home → Close & Load ▼ → Close & Load To...
- 2 Select Only Create Connection → OK.

Day 2 — Merge tables + controls + exception report

Goal: Create and load two tables: Clean_AP_Transactions and Exception_Report. Keep other queries connection-only.

11) Reopen Power Query Editor

- 1 In Excel: Data tab → Queries & Connections.
- 2 Double-click AP_Raw to open Power Query Editor.

12) Import Vendor Alias Map

- 1 Home → New Source → Text/CSV → choose Vendor_Alias_Map.csv → Transform Data.
- 2 Rename query to Vendor_Alias_Map.
- 3 VendorRaw column → Transform → Format → Trim, then Clean.

13) Merge AP_Raw with Vendor Alias (bring VendorName_Clean)

- Click query AP_Raw.
- Home → Merge Queries.
- Top table: Vendor. Bottom: Vendor_Alias_Map → VendorRaw. Join: Left Outer → OK.
- Expand merged column → select only VendorName_Clean → OK.

14) Import Vendor Master

- 1 Home → New Source → Text/CSV → choose Vendor_Master.csv → Transform Data.
- 2 Rename query to Vendor_Master.
- 3 VendorName_Clean column → Transform → Format → Trim, then Clean.

15) Merge Vendor Master (bring VendorID + Category)

- Click AP_Raw.
- Home → Merge Queries.
- Match VendorName_Clean to Vendor_Master VendorName_Clean (Left Outer) → OK.
- Expand merged column → select VendorID and Category → OK.

16) Import Purchase Orders

- 1 Home → New Source → Text/CSV → choose Purchase_Orders.csv → Transform Data.
- 2 Rename query to Purchase_Orders.
- 3 PO_Amount → Transform → Data Type → Decimal Number.
- 4 PO_Number → Transform → Format → Trim.

17) Merge Purchase Orders (bring PO_Amount)

- Click AP_Raw.
- Home → Merge Queries.
- Match PO_Number to Purchase_Orders PO_Number (Left Outer) → OK.
- Expand merged column → select only PO_Amount → OK.

18) Add POMatchStatus (Conditional Column; menu-only)

- AP_Raw selected → Add Column → Conditional Column.
- Name POMatchStatus.
- Rule 1: if PO_Amount equals null → No PO.
- Add Clause: else if Amount equals PO_Amount → Match.
- Else Mismatch → OK.

19) Duplicate counts (use Duplicate query to avoid merge-visibility issues)

- 1 Right-click AP_Raw → Duplicate → rename to DupCounts.

- 2 In DupCounts: select VendorName_Clean and Amount → Home → Remove Columns → Remove Other Columns.
- 3 Home → Group By: group by VendorName_Clean and Amount; Count Rows; name DupCount → OK.

20) Merge DupCount back + add DuplicateFlag

- Click AP_Raw → Home → Merge Queries.
- Select VendorName_Clean and Amount as keys in both AP_Raw and DupCounts (Ctrl/Cmd click). Left Outer → OK.
- Expand merged column → DupCount only → OK.
- Add Column → Conditional Column: DuplicateFlag; if DupCount > 1 then TRUE else FALSE → OK.

21) Rename final AP query to Clean_AP_Transactions

- 1 Right-click AP_Raw → Rename → Clean_AP_Transactions.

22) Create Exception_Report (reference + exception flag + filter)

- Right-click Clean_AP_Transactions → Reference → rename to Exception_Report.
- Exception_Report: Add Column → Conditional Column → name ExceptionFlag.
- Rule 1: VendorName_Clean equals null → Exception.
- Add Clause: POMatchStatus equals Mismatch → Exception.
- Add Clause: DuplicateFlag equals TRUE → Exception.
- Else OK → OK. Filter ExceptionFlag to show only Exception rows.

23) Load ONLY the two outputs to Excel (keep others connection-only)

- Clean_AP_Transactions → Home → Close & Load ▼ → Close & Load To... → Table → New Worksheet → Load.
- Exception_Report → Close & Load To... → Table → New Worksheet → Load.
- In Excel: Data → Queries & Connections → for Vendor_Alias_Map, Vendor_Master, Purchase_Orders, DupCounts: right-click → Load To... → Only Create Connection.

End state (what students should have)

- Clean_AP_Transactions sheet: Amount (Decimal), InvoiceDate (Date), VendorName_Clean + VendorID + Category, PO_Amount, POMatchStatus, DupCount, DuplicateFlag.
- Exception_Report sheet: only rows that are unmapped vendor OR PO mismatch OR duplicate.